Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

Popular Music over time

1. Report on your data management practical experience (not the plan but the application) from the data collection exercise using a minimum of 3-4 sentences (for each item) (5%)
    a. The goal (what it was driven by), and the mode(s) of collection.
        i. We are asking the research question: "**To what extent have the characteristics of popular songs changed over time?"** We are trying to understand the trends in popular music over the years and potentially compare them across ages. We used observation and measurement to watch and track songs over time.
    b. How data were acquired, any problems, lessons learned, etc.
        i. We used 3 datasets: Evolution of Popular Music Using Billboard Hot 100 Data (imported as a JSON file), Billboard Hot 100 & More (imported as a Kaggle dataset), and Spotify Tracks Dataset (imported as a Kaggle dataset). We used Python, specifically the pandas and matplotlib libraries. First, we joined the Billboard Hot 100 & More dataset with the Spotify Tracks Dataset based on the track name and artist name. Next, we joined this with the Evolution of Popular Music Using Billboard Hot 100 Data dataset based on track name, artist, and date. The range of our data is the songs from 1965 to 2025. Once we had this data, we used the matplotlib library to specifically graph the trends of danceability, duration, time, energy, valence, and tempo over the years to discover trends in the data.
        ii. We realized that when we initially joined the datasets, we had only a limited number of songs per year, with the maximum being 42 songs. When debugging the code, we found that the reason a lot of our songs were missing was because the artists were listed differently across the data sets. For example, in one of the datasets the artists could be listed as "Ingrid Michaelson; Zayn" whereas in another dataset it could be listed as "Ingrid Michaelson & Zayn" or just "Ingrid Michaelson". This causes issues when joining and these songs and artists get dropped. We were able to fix this and get back more of the eligible songs.
        iii. We learned that it is important to be careful when handling the data because something as small as the artists being listed differently caused us to lose most of the data. It is important to match up the data columns and see how the values are stored before joining datasets together. This would've skewed our data and our analysis.
    c. Physical and logical organization of data/ metadata, successes, failures.
        i. We imported the data and joined the data based on track name, artist and date. This made it easy to identify a particular track and gave us the 3 key data points needed to identify a song. The readability of the data is fairly simple because the column names are listed above and are

Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

Popular Music over time

self-explanatory. We created a public repository on GitHub and committed the code there, so that it is accessible to anyone that wants to use it.

d. Metadata and documentation collected/ stored, what provenance was identified.
   i. Our metadata is primarily based on background information about the songs themselves. We are taking into account the trends of danceability, duration, time, energy, valence, and tempo over the years to discover trends in the data. All of these are metadata about the data (the songs themselves) because it gives us more information into the details of the song and how they are different from other songs. The metadata comes from the Kaggle Datasets and the imported JSON files, and we are able to trace the origins of these sources because we know the authors and also the reliability of these sources.
e. A link to the data/ metadata (or a copy of it)
   i. We wrote and executed the code in Google Colab and then committed the code to the GitHub repository. This is the link to our public GitHub repository: https://github.com/sreelasyagogineni/Evolution-Of-Popular-Music

2. Describe how well (or not) the data management plan turned out in practice. 6% (6000-Level students)/ 8% (4000-Level, students)
   a. Highlight what worked, what did not, and why (2-3 sentences for each of the 9 data management categories).
      i. Creation of logical collections
         1. We said that we would primarily organize the data by year, but we ended up organizing the data by track, artist and date. We had to do it this way because an artist can have many tracks in a year, so we need to have the track, artist and year to verify that it is unique. We said that our main columns will be year, artist, genre, duration, BPM, and chart positions. However, we ended up organizing by metadata of the song like danceability, duration, time, energy, valence, and tempo because that is what we are using for our analysis.
      ii. Physical data handling
         1. We used the same datasets as mentioned in our data management plan. A GitHub repository called Historic Billboard Hot 100 Data also provided the exportable JSON files that can be used to extract data. Billboard Hot 100 & More was imported as a Kaggle dataset; Spotify Tracks Dataset was imported as a Kaggle dataset. This data was accurate and allowed us to find and organize the metadata for each song as needed.

Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

Popular Music over time

iii. Interoperability support
    1. All of our datasets are available on any system since they're all CSVs or JSON files. Because the GitHub repository and Kaggle are both open source, anyone can access them. We also made our cleaned dataset available on our GitHub Repo along with our Colab Notebook code.

iv. Security support
    1. This is the same as what we initially planned for in our data management plan because we carefully analyzed the security of the datasets before using them. Regarding security, neither the repository nor Kaggle is serving any personal information. Confidentiality is not a problem, as both are open source. Additionally, GitHub has its own security logs to track data vulnerabilities. The public, including us, do not have access to the actual data collected; it is just openly available.

v. Data ownership
    1. This is the same as what we initially planned for in our data management plan because we carefully analyzed the data ownership for each of the datasets before using them. We were able to join the datasets and ensured that the years were aligned and specifically used a time frame of 1965 to 2025 so that the data is coming from both data sets. One of the datasets started collecting data in 1958 while the other datasets started collecting data in 1963, so we started collecting the data from 1965.

vi. Metadata collection, management, and access
    1. In terms of metadata collection, we mostly stuck to the initial data management plan. However, we ended up focusing more on the metadata and had those as their own columns. We realized during the process of collecting data that these are the factors that will help us answer our research question, so we factored that in more when we were joining the datasets to create our final dataset.

vii. Persistence
    1. For persistence, we stuck with what we wrote in our initial data management plan. We added the links to all our the datasets: Evolution of Popular Music Using Billboard Hot 100 Data (imported as a JSON file), Billboard Hot 100 & More (imported as a Kaggle dataset), and Spotify Tracks Dataset (imported as a Kaggle dataset). We are adding our findings and conclusions in a

Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

Popular Music over time

GitHub repository so people who want to learn more can access them.

viii. Knowledge and information discovery
   1. As mentioned in persistence, all the datasets (The Kaggle datasets, The Billboard Hot 100 and Spotify Tracks Dataset) we used to create our joined dataset will be added to our repository. In addition, we are adding the code for our joined dataset to our public GitHub repository. Furthermore, we will be adding our findings and conclusions so that people can access them and replicate this study if they want to.

ix. Data distribution and publication
   1. For this, we stuck with what we wrote in the initial data management plan. The data we use is the Billboard Hot 100 Rankings, which is updated every week. We also use the GitHub repository, which is updated weekly. For the metadata, we are using the Spotify Tracks dataset, and we will also add the link so that the data is accessible. Since our joined dataset uses the data from the datasets that are continuously updated, our project will continue to be up to date even after we finish this project. The graphs and tables may update, so external viewers may run the python Notebook to generate expanded findings.

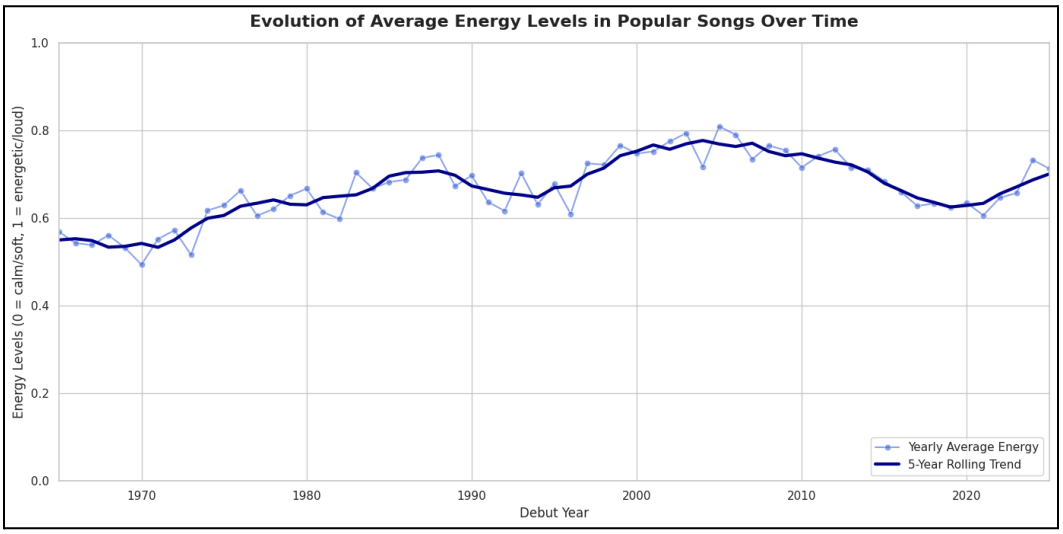
b. What would you do differently next time and why? (2-3 sentences total)
   i. Next time we would also like to collect data from individuals on their answer to our research question: "**To what extent have the characteristics of popular songs changed over time?",** and give them some options on trends over the years. Using this data, we can compare how people's responses compare against the actual data by graphing both. This would be interesting to see because it would be nice to compare public opinion on popular songs vs what the data actually shows.

c. Any insights about the data or metadata collected: completeness, quality, etc.? (2-3 sentences total)
   i. We believe that the data and metadata collected are accurate because we used multiple data sources to ensure that the data was cross checked. Before using these sources, we also ensured that the origins of these sources were trustworthy and reliable. We combined the data of different datasets to get the bigger picture and more metadata on each song.

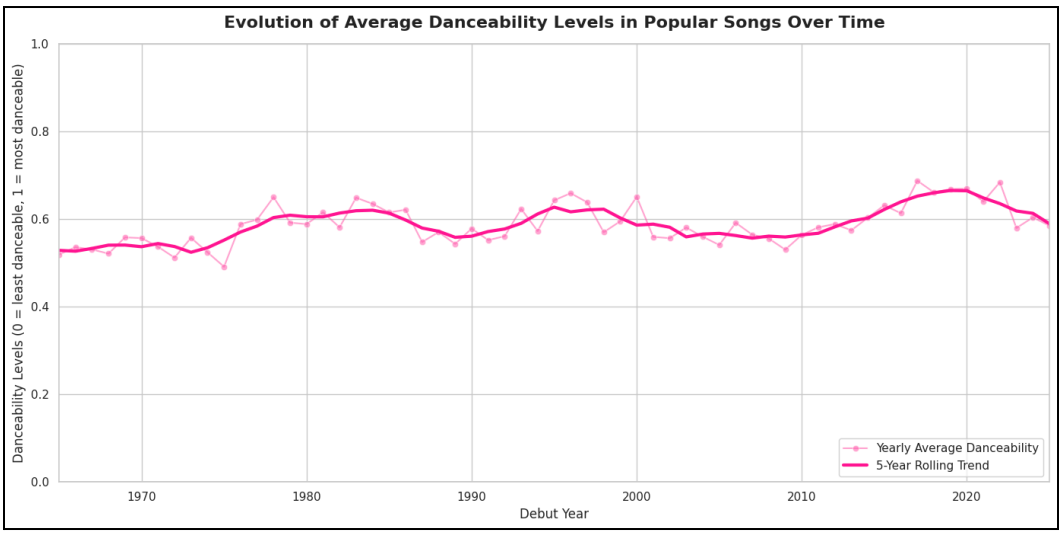
3. Conduct some preliminary/exploratory descriptive analysis of a feature (or more) in your dataset. 2%
   a. Energy Levels vs. Time

Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

Popular Music over time



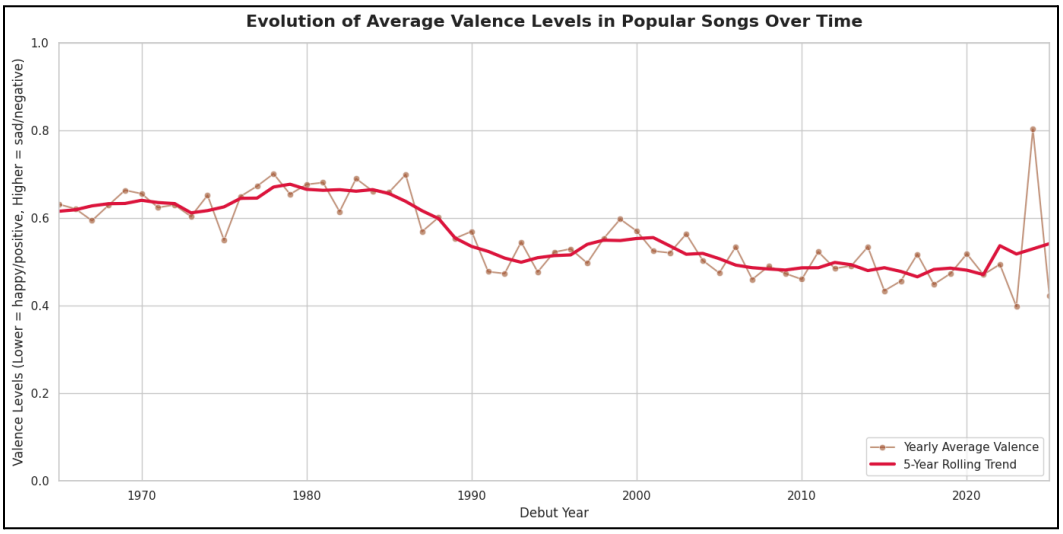
b. Danceability vs. Time



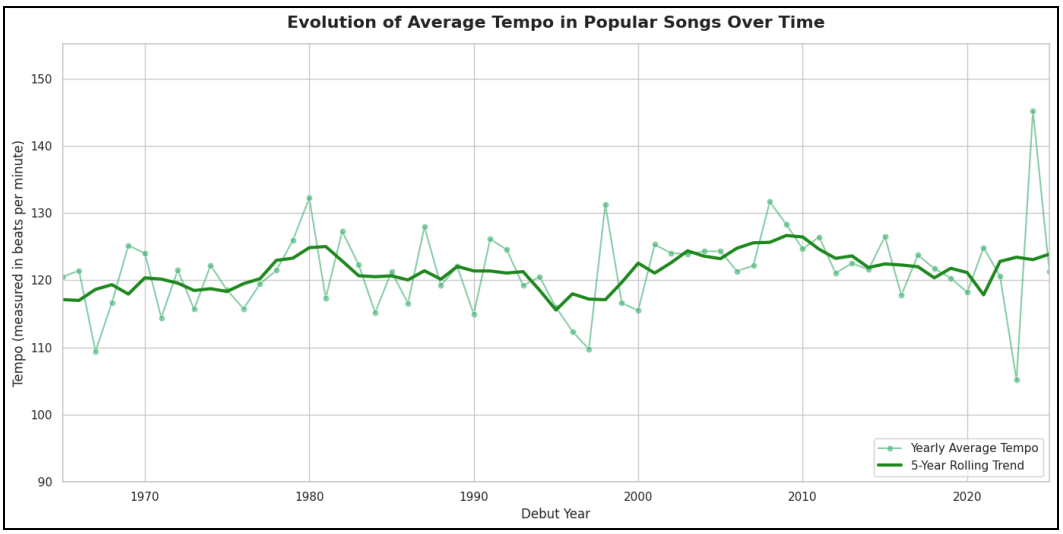
c. Valence vs. Time

Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

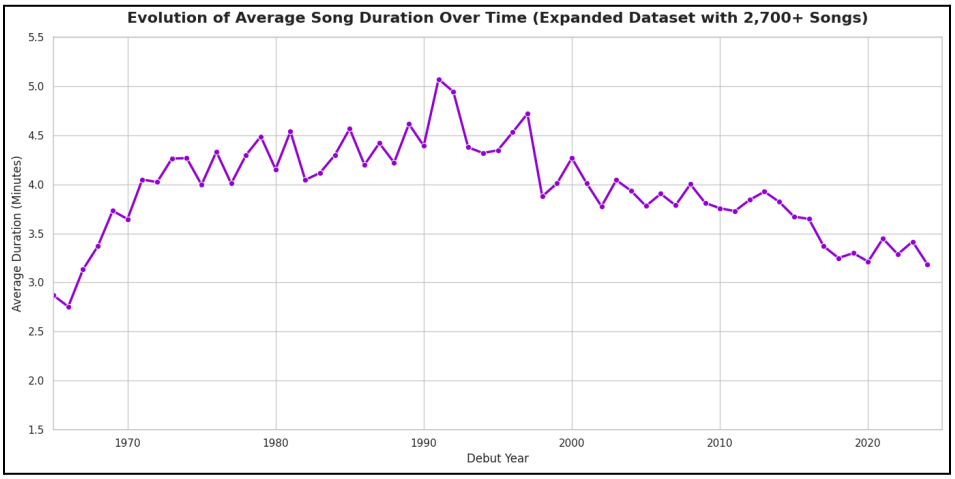
Popular Music over time



d. Tempo vs Time



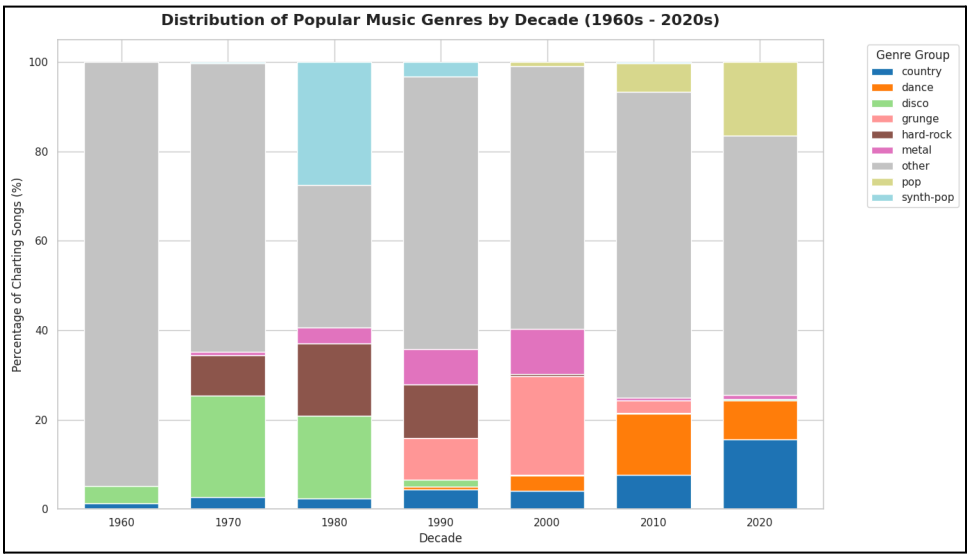
e. Duration vs Time



Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

Popular Music over time

f. Genre By The Years



4. 6000-Level student question: related to Q 1 and Q 2. 2% — **ananya only**
   a. What standards (data format, metadata, etc.) did you use (whether planned or not) and/or what ones couldn't you use and why? (2-3 sentences).
      i. We used a relational table with the pandas library to join all three datasets and for grouping the rows by year. To merge the data, we had to make all the data points lowercase, strip the word of any spaces, and take out subtle formatting differences. We created dedicated tracking columns such as *clean_title,* and *clean_artist*. Additionally, we standardized the dates into ISO date formatted (YYYY-MM-DD) when merging the Github Hot 100 dataset with the Spotify dataset. Normalizing columns such as *danceability, energy, and valence* was necessary because we wanted the data to be more uniform and allow for direct comparisons without scaling mismatches.
      ii. We could not use exact joins with primary keys. When we relied on matching the original columns (track_name and artists), we missed data points that included commas, semicolons, and artist designations (feat., with, etc).
   b. What standards/conventions would have been more helpful ('none' is not an acceptable answer) ? Minimum 1 sentence.
      i. It would have been more helpful if these three datasets used the same id system such as having a standard IRSC (International Standard Recording Code). Then, we would have had to parse through them and use complicated regex logic.

Ananya Subramaniam
Sree Lasya Gogineni
Data Science F26

Popular Music over time

c. How do your choices of organization, formats, frameworks, and standards help capture and elucidate the provenance of your data product(s)? (3-4 sentences).
   i. We used an interactive Python Notebook that we organized linearly by loading the three datasets individually. After parsing and joining the data, we had to debug it which led to us being able to convert the data into visualizations. With this notebook, an external user can trace back the dataframes and their origin. We also preserved the raw CSV files to have the ground truth be accessible. Additionally, we created multiple copies of the dataframe to manipulate throughout the joining process.